36 Programming and DS (12)

36.1 Algorithm Design (1)

36.1.1 Algorithm Design: NIELIT 2021 Dec Scientist B - Section B: 109

What does the following code do ?

```
int a, b;
a = a + b;
b = a − b;
a = a − b;
```

A. leaves a and b unchanged
B. a doubles and stores in a
C. b doubles and stores in a
D. Exchanges a and b

nielit2021dec-scientistb programming-in-c algorithm-design output

Answer key☟

36.2 Compiler tokenization (2)

36.2.1 Compiler tokenization: NIELIT 2022 April Scientist B | Section B | Question: 70

What does the following C-statement declare?

```
int (*f) (int * );
```

A. A function that takes an integer pointer as argument and returns an integer.

B. A function that takes an integer as argument and returns an integer pointer.

C. A pointer to a function that takes an integer pointer as argument and returns an integer.

D. A function that takes an integer pointer as argument and returns a function pointer.

nielit2022apr-scientistb programming-in-c pointers compiler-tokenization

36.2.2 Compiler tokenization: NIELIT Scientist B 2020 November: 47

The number of tokens in the following C statement is

```
printf("i=%d, &i=%x", i, &i);
```

A. 8 B. 4 C. 7 D. 10

nielit-scb-2020 compiler-tokenization programming-in-c

Answer key☟

36.3 Functions (1)

36.3.1 Functions: NIELIT 2021 Dec Scientist B - Section B: 67

Choose correct statement for code segment

```
int multiply (int a, int b = 5),
```

A. variable a and b are of int types and the initial value of a and b are 5
B. variable b is of int type and will always have value 5
C. variable b is of global scope and will have value 5
D. variable b will have value 5 if not specified when calling a function in further program

nielit-2021-it-dec-scientistb programming-in-c functions

36.4 Memory Management (1)

36.4.1 Memory Management: NIELIT 2021 Dec Scientist B - Section B: 26

In $C++$, dynamic memory allocation is accomplished with which of the following operator:

A. this
B. new
C. delete
D. malloc()

nielit-2021-it-dec-scientistb programming-in-c data-structures memory-management

Answer key☟

36.5 Output (1)

36.5.1 Output: NIELIT 2021 Dec Scientist B - Section B: 34

Output of following code

```
main(){printf ("\n%%%%");}
```

A. Error message
B. $\%\%$
C. 00
D. $\&\&\&\&$

nielit-2021-it-dec-scientistb programming-in-c output

Answer key☟

36.6 Parameter Passing (1)

36.6.1 Parameter Passing: NIELIT 2022 April Scientist B | Section B | Question: 76

What is printed by the print statements in the program $P1$ assuming call by reference parameter passing ?

```
Program P1()
{
   x = 10;
   y = 3;
   func1(y, x, x);
```

```
    print x;
    print y;
}
func1(x, y, z)
{
    y = y+4;
    z = x+ y+ z;
}
```

A. $10, 3$ B. $31, 3$ C. $27, 7$ D. $33, 5$

nielit2022apr-scientistb   programming-in-c   parameter-passing   data-structures

Answer key☟

36.7 Pointers (1)

36.7.1 Pointers: NIELIT 2022 April Scientist B | Section B | Question: 95

Consider the following C code:

```
#include<stdio.h>
int *assignval (int *x, int val) {
    *x = val;
    return x;
}

int main () {
    int *x = malloc(sizeof(int));
    if (NULL == x) return;
    x = assignval (x,0);
    if (x) {
        x = (int *)malloc(sizeof(int));
        if (NULL == x) return;
        x = assignval (x,10);
    }
    printf("%d\n", *x);
    free(x);
}
```

The code suffers from which one of the following problems:

A. compiler error as the return of malloc is not typecast appropriately.
B. compiler error because the comparison should be made as $x == \text{NULL}$ and not as shown.
C. compiles successfully but execution may result in dangling pointer.
D. compiles successfully but execution may result in memory leak.

nielit2022apr-scientistb   programming-in-c   memory-management   pointers

36.8 Programming In C (4)

36.8.1 Programming In C: NIELIT 2021 Dec Scientist B - Section B: 49

The best statement to find out that value of a variable lies in the range 7 to 10 or 12 to 15 in C language is :

A. switch B. while C. for D. continue

nielit-2021-it-dec-scientistb programming-in-c

Answer key

36.8.2 Programming In C: NIELIT 2021 Dec Scientist B - Section B: 55

In C language value of $\mathrm{ix}+\mathrm{j}$, if j is of integer type and ix long type would be:

A. Integer B. Float C. Long integer D. Double precision

nielit-2021-it-dec-scientistb programming-in-c

36.8.3 Programming In C: NIELIT 2021 Dec Scientist B - Section B: 65

If $\mathrm{a}{+}{+}$ is replaced with ${+}{+}\mathrm{a}$, which statement does not get affected?

A.
```
printf("% d % d", ++a, a++);
```
B.
```
a = 20; a++;
```
C.
```
while(a++ = 20) cout<<a;
```
D.
```
d = a++;
```

nielit-2021-it-dec-scientistb nielit2021dec-scientistb programming-in-c

36.8.4 Programming In C: NIELIT 2021 Dec Scientist B - Section B: 91

Following are implicitly provided in C programming language :

A. Output facility
B. Input facility
C. Both Input and Output facility
D. No Input and Output facility

nielit2021dec-scientistb programming-in-c

Answer Keys

36.1.1	D	36.2.1	C	36.2.2	D	36.3.1	D	36.4.1	B
36.5.1	B	36.6.1	B	36.7.1	D	36.8.1	A	36.8.2	C
36.8.3	B	36.8.4	C						